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Research Article

Efficient Isolation of Cellulosic Fibers from Coffee Parchment via Natural Acidic Deep Eutectic Solvent Pretreatment for Nanocellulose Production

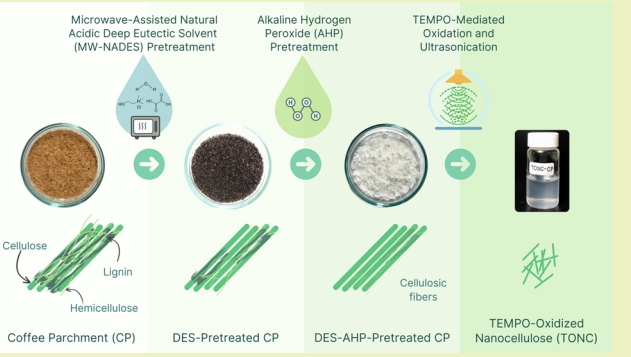
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ABSTRACT: The efficient pretreatment of lignocellulosic biomass is essential for cellulose valorization. This work investigated the fractionation efficiency of the pretreatment process based on natural acidic deep eutectic solvent (NADES) for isolating cellulose-rich materials (CRM) from coffee parchment (CP). The results showed that NADES, composed of choline chloride, oxalic acid, and water in a 2:2:1 molar ratio, was the most efficient solvent for CP fractionation. The sequential pretreatment using NADES under microwave heating (MW-NADES) at 130 °C for 15 min, followed by a two-step alkaline hydrogen peroxide (AHP) process, could maximize the removals of hemicellulose (99.6%) and lignin (97.7%) from CP, leading to production of the CRM with 86.1% cellulose content. This process generated a rough, fractured surface and promoted microfibril disassembly observed in the pretreated fibers. As a result, the CRM exhibited a highly crystalline structure and good thermal stability. Furthermore, the results also highlighted the successful transformation of the isolated CRM to nanofibril networks having a diameter of less than 10 nm and a length of several hundred nanometers. Overall, this work suggests that the sequential MW-NADES and AHP pretreatment could be considered an effective approach in isolating high-purity cellulose from CP, potentially a sustainable source for nanocellulose production.



KEYWORDS: *coffee parchment, biomass pretreatment, deep eutectic solvents, alkaline hydrogen peroxide, microwave-assisted extraction, nanocellulose*

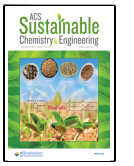
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INTRODUCTION

Nanocellulose is a term referring to a material composed primarily of cellulose, having at least one dimension in nanoscale.[1] Currently, it has emerged as a promising sustainable biomaterial for biomedical and pharmaceutical applications such as drug delivery systems, tissue engineering scaffolds, reinforced composites, wound dressings, and biosensing devices due to its renewability, biocompatibility, biodegradability, lightweight, high surface area, high mechanical strength, tunable surface chemistry, low thermal expansion, and low cytotoxicity.[2] Based on the isolation method and its native origin, nanocellulose can be typically classified into three main types, including cellulose nanocrystals (CNCs), cellulose nanofibers (CNFs), and bacterial nanocelluloses (BNCs).[3] Regarding plant-derived nanocellulose, CNCs are rod-like or whisker-shaped particles 3 to 50 nm in width and 100 to 500 nm in length, which are typically produced by acid hydrolysis. CNFs are long, flexible chains containing crystalline and amorphous regions with a cross-sectional diameter of 10 to 100 nm and a length of 1 to 10 μm, which are commonly obtained by mechanical disintegration.[1,2]

One of the most abundantly available and renewable sources of cellulose fibers is agro-industrial byproducts or agricultural residues.[4,5] In Thailand, arabica coffee has become one of the main economic crops mostly cultivated in the highlands of the northern region, particularly Chiang Mai Province. The production of arabica coffee in northern Thailand is about 10,000 tons each year.[6] The waste generation is an inevitable consequence of coffee processing. Coffee parchment (CP) is a pre-roasting yellowish byproduct generated during the drying and hulling steps in the wet processing of the coffee cherry. The parchment of coffee is also known as an endocarp that covers both hemispheres of the coffee seed, representing 3.5−6.1% dry weight of coffee cherry. Most of this biomass residue is unutilized and usually managed by open-field burning.[7]

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According to the literature, CP has a strong fibrous tissue containing 35−54% of cellulose, 20−38% of hemicellulose, and 27−35% of lignin, with low amounts of ash, lipids, and proteins.[8−11] Due to its high insoluble fiber content, CP has been reported as a reinforcement material in polymer composites and dietary fiber ingredients.[12,13] However, the high lignin content of CP in nature provides the strong lignin-carbohydrate complexes in biomass, making it very resistant to chemical and enzymatic degradation, thus hindering the successful conversion of lignocellulose into value-added products.[14]

In order to isolate cellulose fibers from biomass, pretreatment is a critical step required to overcome the biomass recalcitrance by altering cell wall features: lignin sheath breakdown, hemicellulose dissociation, and reduced crystallinity/depolymerization of cellulose.[4,15] Traditional chemical pretreatment methods, including acid hydrolysis, alkali treatment, organosolv process, steam explosion, and kraft pulping process, have high efficiency in removing hemicellulose and lignin. However, these conventional pretreatments are usually carried out at very high temperatures and long reaction times, causing issues of hazardous waste generation, high energy consumption, and toxic inhibitor formation.[16−18] Reis et al. reported the production of cellulose microfibers from CP using a three-step method: alkaline treatment (2% NaOH solution) at 120 °C for 1 h, followed by a steam explosion at 120 °C, 2 bar for a total of 45 min and high shearing homogenization process at 8000 rpm for 1 h.[12] Nevertheless, many studies have demonstrated the utilization of lignocellulose from CP without any fractionation.[13,19−21]

Therefore, it is important to establish efficient and environmentally friendly pretreatment approaches for lignocellulose fractionation. Deep eutectic solvent (DES), a new class of ionic liquids, is defined as a mixture of two or more components at a certain proportion with a melting point much lower than that of individual constituents.[22] Basically, the formation of DES is predominately associated with hydrogen bonding interactions between hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD).[23] Recently, the term "natural DES" has emerged as a green solvent obtained by natural primary metabolites such as sugars, organic acids, amino acids, and choline derivatives.[24] In general, DESs share many characteristics with ionic liquids, including nonvolatility, nonflammability, recyclability, high tunability, and being liquids at a wide range of temperatures. However, most natural DESs have potentially minimum environmental impacts with low toxicity and great biodegradability compared with conventional ionic liquids and hazardous organic solvents.[25] The physicochemical properties of natural DESs, such as viscosity, conductivity, and polarity, can be tailored by altering HBA and HBD molecules and their molar ratios or by the addition of water, making them attractive extraction solvents in biomass pretreatments.[26] Numerous publications have reported the great efficiency of natural acidic DESs (or NADESs) in removing hemicellulose and lignin from various biomass feedstocks such as chestnut shell,[27] rice straw,[28] poplar wood flour,[29] and corncob.[30] The study of Hou et al. demonstrated that choline chloride-oxalic acid NADESs in a 2:1 molar ratio refluxing at 120 °C for 1 h could remove hemicellulose up to 99.6% and lignin up to 75.9% from rice straw.[28]

Herein, this research aimed to develop an effective pretreatment strategy based on the use of NADES combined with microwave-assisted extraction, called MW-NADES, for isolating high-purity cellulose from CP as a potential source for nanocellulose production. The effects of molar compositions of the DES system, microwave temperature, and the sequential pretreatment using MW-NADES and alkaline hydrogen peroxide (AHP) bleaching process on biomass fractionation efficiency were investigated based on the chemical compositions including cellulose content, hemicellulose removal, and delignification. The changes in chemical structure, crystallinity index, morphology, and thermal stability upon pretreatments were investigated using Fourier transform-infrared (FT-IR) spectroscopy, X-ray diffraction (XRD), scanning electron microscopic (SEM), and thermogravimetric analysis (TGA), respectively. Furthermore, the isolated cellulosic fibers were then transformed to nanocellulose.

EXPERIMENTAL SECTION

Raw Material and Chemicals. CP, a byproduct of *Coffea arabica* L., was received from a traditional farm in Mae Kampong Village, Mae On District, Chiang Mai Province, Thailand, in October 2020. The biomass feedstock was air-dried, mechanically ground to 20−80 mesh particle size, and oven-dried at 45 °C until a constant weight was achieved and stored in an airtight container before pretreatment. Choline chloride (BioReagent, ≥98%), oxalic acid dihydrate (ACS reagent, ≥99.5%), hydrogen peroxide (stabilized for synthesis, 30%), 2,2,6,6-tetramethylpiperidine-1-oxyl radical (TEMPO; 98%), and sodium hypochlorite (reagent grade, 4.99% available chlorine) were purchased from Sigma-Aldrich (St. Louis, MO, USA). Commercial CNC (4 to 10 nm in width, 100 to 260 nm in length) was received from Cellulose Lab (NB, Canada). Pharmaceutical secondary standards of D-(+)-glucose and D-(+)-xylose were purchased from Sigma-Aldrich (St. Louis, MO, USA). All other chemicals were of analytical or HPLC grade and used without purification.

NADES Preparation. In this study, a system of NADES was developed by constructing a ternary phase diagram of choline chloride (ChCl), oxalic acid (OA), and water. All mixtures were prepared by the heating and stirring method. Before mixing, pure ChCl and OA were dried at 40 °C in a vacuum oven for 48 h to remove free water. Afterward, ChCl and OA were weighed and transferred to glass vials which were subsequently added with deionized (DI) water at the desired molar ratios (the initial amounts of ChCl and OA were 5 mol % for each). The mixtures were heated at 80 °C with continuous stirring until a homogeneous transparent liquid was formed. The resulting liquids were kept at room temperature in sealed vials before use. Twelve NADESs were selected for further pretreatments, as listed in Table 1. The viscosity and Kamlet−Taft parameters of selected NADESs were characterized, as described in the Supporting Information.

MW-NADES Pretreatment. In brief, 1 g of dried raw CP was dispersed in 12 selected NADESs in a modified polytetrafluoroethylene (TFM) reaction vessel (JK-100, Sineo, Shanghai, China) with a biomass-to-solvent ratio of 1:10. The vessels were then placed in a rotating carousel equipped with a high-precision platinum resistance temperature sensor to control a microwave temperature precisely. The pretreatments were all performed at 100 to 130 °C for 15 min using a Jupiter-B closed-vessel microwave extraction system (Sineo, Shanghai, China), having a frequency of 2450 MHz with a fixed power output of 1000 W. At the end of pretreatment, the reaction was terminated by adding 20 mL of ethanol/water mixture (1:1, v/v) as an anti-solvent. The solid fractions were separated from liquids via vacuum filtration and continuously washed with an anti-solvent, followed by DI water. The solid residues were oven-dried at 60 °C to a constant weight and stored in a desiccator for further use. All pretreatments were performed in duplicates. All pretreated solids were designated as DES-CP.

AHP Pretreatment. After MW-NADES pretreatment, the solid residues were sequentially bleached with an AHP solution. The AHP process was performed according to a reported method with slight

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Table 1. Twelve Selected Mixtures of ChCl−OA−Water System with Various Molar Compositions

NADESs	eutectic compositions (mole fraction)		
	x_{ChCl}	x_{OA}	x_{water}
DES-01	0.10	0.05	0.85
DES-02	0.20	0.10	0.70
DES-03	0.30	0.15	0.55
DES-04	0.40	0.10	0.50
DES-05	0.40	0.20	0.40
DES-06	0.40	0.40	0.20
DES-07	0.45	0.45	0.10
DES-08	0.50	0.25	0.25
DES-09	0.50	0.50	
DES-10	0.60	0.20	0.20
DES-11	0.60	0.30	0.10
DES-12	0.67	0.33	

modifications.[31] The dried DES-CP samples were soaked in a 5% w/v hydrogen peroxide (H_2O_2) solution, equivalent to 1000 mg H_2O_2 per g of dried biomass, at pH 11.5 ± 0.1, with a 10% w/v biomass loading. The biomass suspensions were subsequently agitated using an LSI-3016R shaking incubator (Daihan Labtech, Korea) at 120 rpm, 60 °C for 24 h. After that, the pretreated samples were filtered and thoroughly washed with DI water to neutral pH. The solid fractions were collected, oven-dried at 60 °C, and stored in a desiccator before characterization. The solid residues obtained from the sequential MW-NADES followed by the bleaching process were denoted as DES-AHP CP, later called cellulosic fibers or cellulose-rich material (CRM). The AHP pretreatments were carried out in duplicates.

Nanocellulose Production. The nanocellulose was prepared by TEMPO-mediated oxidation, followed by ultrasonication.[32−35] Typically, 1 g of CRM was dispersed in 100 mL of DI water containing 0.016 g of TEMPO and 0.1 g of sodium bromide. The catalytic oxidation was initiated by adding sodium hypochlorite (NaClO) solution (10 mmol per gram of cellulose) to the cellulose suspension at room temperature under continuous stirring. The pH of the suspension was maintained at 10.0 ± 0.1 by adding 0.5 M NaOH solution. After 4 h, the reaction was terminated with ethanol, and the slurry was washed with DI water by centrifugation until the pH of the supernatant reached neutral. The solid fraction was named TEMPO-oxidized cellulose fiber (TOCF). To produce nanostructured cellulose, TOCF was then mechanically disintegrated using a JY88-IIN ultrasonic homogenizer (Drawell, China) equipped with a 6 mm accompanying tip. The ultrasonication process was carried out at an output frequency of 20 to 25 kHz and power of 250 W (75% duty ratio) with 10 min/cycle for 5 cycles. The homogeneous, gel-like substance produced was called TEMPO-oxidized nanocellulose (TONC-CP). The resulting nanocellulose was stored at 4 °C without drying.

Determination of Structural Carbohydrates and Lignin. The chemical composition of untreated and pretreated CPs was analyzed according to the National Renewable Energy Laboratory (NREL/TP-510-42618) standard protocol.[36] In brief, untreated and pretreated samples were pre-extracted with water and ethanol to remove any soluble extractives before a two-step acid hydrolysis. Then, the samples were hydrolyzed with 72% sulfuric acid at 30 °C for 60 min, followed by 4% sulfuric acid at 121 °C for another 60 min. The hydrolysates were then filtered under a vacuum. Acid-insoluble lignin (AIL) was determined using the gravimetric method after drying to constant weight at 105 °C. In contrast, acid-soluble lignin (ASL) was analyzed by measuring the UV−vis absorbance of the filtrates at 240 nm using a UV2600i UV−vis spectrophotometer (Shimadzu, Japan). The total lignin content comprised the amounts of AIL and ASL. Monosaccharides in the filtrates were identified and quantified using an Agilent 1260 Infinity II high-performance liquid chromatography (Agilent Technologies, CA, USA), equipped with a refractive index detector and a SUPELCO C−610H column (300 mm length and 7.8 mm inner diameter; Sigma-Aldrich, PA, USA). The temperatures of the column and detector were maintained at 50 °C. The diluted samples with 10 μL injection volume were eluted with 5 mM sulfuric acid at a 0.6 mL/min flow rate. Glucose and xylose were used as analytical standards. The cellulose content was calculated as glucan, converted from glucose, while the major hemicellulose found in untreated CP was xylan, calculated from xylose. The percentages of solid recovery, cellulose recovery, cellulose removal, hemicellulose removal, and lignin removal (delignification) were calculated using the equations below

$$\%\ \text{solid recovery} = \frac{\text{residual mass of sample after pretreatment}}{\text{initial mass of untreated sample}} \times 100 \quad (1)$$

$$\%\ \text{component recovery} = \frac{m_2}{m_1} \times 100 \quad (2)$$

$$\%\ \text{component removal} = \frac{m_1 - m_2}{m_1} \times 100 \quad (3)$$

where m_1 is the content of individual lignocellulose components present in untreated CP and m_2 is the content of the component remaining in the sample after pretreatment.

FT-IR Spectroscopy. The FT-IR spectra for untreated CP, pretreated CP, and nanocellulose were acquired using a Nicolet 6700 FT-IR spectrometer (Thermo Scientific, WI, USA) equipped with a single-bounce attenuated total reflectance (ATR) with a diamond crystal. The dried samples were examined in absorbance mode over 400 to 4000 cm^{-1} with 32 scans per spectrum and 6 cm^{-1} resolution.

X-ray Diffraction. The XRD measurements for untreated and pretreated CPs were conducted using a D8 Discover X-ray diffractometer (Bruker AXS, Germany) equipped with a Göbel mirror. The dried samples were continuously scanned with Cu Kα radiation (λ = 0.154 nm) operated at 40 kV and 40 mA. The scattered radiation was detected in the diffraction angle 2θ ranging from 5 to 60°, with a scan rate of 0.8 s per step and a step increment of 0.02°. The degree of crystallinity or crystallinity index (CrI) of each sample was calculated from the XRD patterns using the empirical equation according to Segal's method,[37] as described below

$$\text{CrI}\ (\%) = \left(\frac{I_{200} - I_{AM}}{I_{200}}\right) \times 100 \quad (4)$$

where I_{200} is the peak intensity of the (200) lattice plane of cellulose crystals located at 2θ = 22.6° and I_{AM} is the intensity of the amorphous diffraction peak at 2θ = 18.0°.

Scanning Electron Microscopy. The surface morphological characteristics of untreated and pretreated CPs were investigated using a Tescan Clara ultrahigh resolution scanning electron microscope (Tescan, Czech Republic). Before observing, the dried samples were mounted on an aluminum stub with conductive carbon tape and sputter-coated with gold. The SEM imaging was operated at an accelerating voltage of 15 kV.

Thermal Analysis. Thermogravimetry (TG) was carried out using a Thermo plus EVO2 simultaneous thermal analyzer (Rigaku, Japan). The dried untreated and pretreated CPs were heated from room temperature up to 500 °C at a heating rate of 10 °C/min under nitrogen purging at a flow rate of 80 mL/min.

Transmission Electron Microscopy. The topological structures of individual nanocellulose were imaged using a JEM-1400 transmission electron microscope (JEOL, Tokyo, Japan). The TONC-CP dispersion (0.25 wt %) was dropped on a carbon-coated copper grid, dried at ambient temperature, and negatively stained with 1% w/v uranyl acetate. The transmission electron microscopy (TEM) imaging was taken at an accelerating voltage of 80 kV. The width and length of individual particles were measured from the TEM images using ImageJ processing software.

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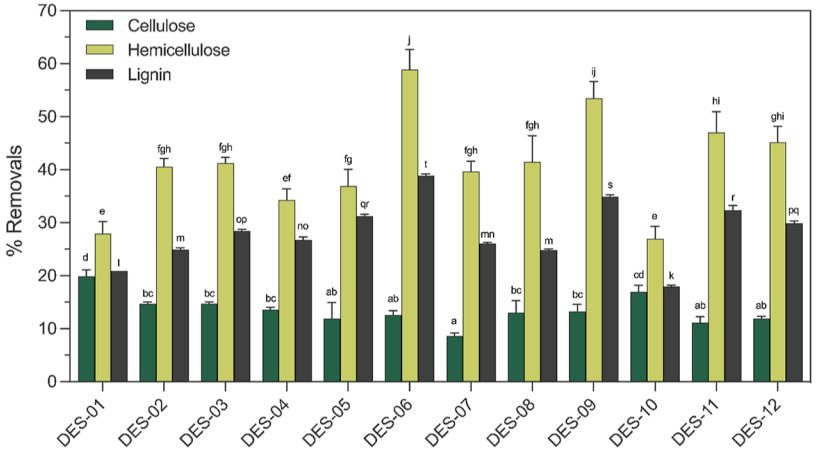


Figure 1. Percent removals of individual lignocellulosic components from CP after the MW-NADES pretreatment at 100 °C for 15 min. All data (presented as mean ± standard deviation) were analyzed using IBM SPSS Statistics, Version 17.0. Values annotated with different lowercase letters (a−d for cellulose; e−j for hemicellulose; k−t for lignin) represent the significant differences (One-Way ANOVA, Tukey's test: $p < 0.05$).

RESULTS AND DISCUSSION

Phase Behavior of Deep Eutectic System. The NADESs used in this study were simply prepared by mixing ChCl (as an HBA molecule) with OA (as an HBD molecule) and water. In order to identify a uniform composition, a ternary phase diagram of the ChCl−OA−water system was constructed as a function of molar composition. The possible phases of the ChCl−OA−water system are graphically represented in Figure S1. The phase diagram separates the physical phase behaviors of the ChCl−OA−water system into single-phase liquid and two-phase solid−liquid mixtures. In the absence of water, a stable liquid phase of binary mixtures was observed when the molar ratios of ChCl to OA were in the range of 1:1 to 7:3. The DES formation at these HBA/HBD stoichiometric ratios was involved by establishing hydrogen bonding interactions between chloride anions of ChCl and hydroxyl groups of carboxylic acid.[38,39] Conversely, phase separation occurred when the ChCl/OA ratio deviated from this range (particularly $x_{OA} > 0.5$). The precipitated phase could arise from excess unreacted molecules, resulting in heterogeneous mixtures.

Furthermore, the phase diagram also reveals that the ChCl−OA mixtures were miscible with water at specific molar ratios to form hydrated DESs, as represented by the green dots within a green area. Adding water could expand the ChCl/OA ratio range to 9:1 without phase separation. However, the water miscibility of ChCl/OA mixtures decreased with the increment of OA mole fraction. The results showed that water content was limited at 20 mol % or 3.28 wt % in a 1:1 ChCl/OA mixture, while it could increase to 85 mol % or 43.04 wt % in a 2:1 ChCl/OA mixture. The low water absorption capability might be attributed to the strengthened nanostructure of DESs caused by the increased number of hydrogen bonds between ChCl and OA.[40−42] The strong intermolecular forces in the DES nanostructure induced resistance of a liquid to flow, which could be explained by the solvent viscosity, as shown in Tables S1 and S2. At all tested temperatures, the viscosity values of a 1:1 ChCl/OA mixture (DES-09) were relatively higher than those of a 2:1 ChCl/OA mixture (DES-12). According to the Arrhenius equation, the activation energy (E_η) of DES-09 (57.6 kJ·mol^{-1}) was greater than that of DES-12 (45.6 kJ·mol^{-1}), suggesting the stronger intermolecular forces in ChCl/OA (1:1) than in ChCl/OA (2:1).[43,44] On the other hand, higher water contents can weaken the hydrogen bonding interactions between HBA and HBD, thus completely dissociating the DES into hydrated ions.[40,41,45]

Effect of Molar Composition of the ChCl−OA−Water System on Fractionation Efficiency of the MW-NADES Pretreatment. On the basis of homogeneous liquid, 12 NADESs with different molar compositions were selected for being investigated in their pretreatment efficiency, as listed in Table 1. In this investigation, the NADES pretreatments were all carried out using microwave irradiation at 100 °C for 15 min, the conditions obtained from our preliminary study. The results demonstrated a great fractionation efficiency in the NADESs with a high OA-to-ChCl ratio (Figure 1). Regarding the absence of water, the hemicellulose and lignin removals increased from 45.1 to 53.4% and 29.9 to 34.8%, respectively, when the OA/ChCl ratio increased from 1:2 (DES-12) to 1:1 (DES-09). The proton-catalyzed cleavages of chemical linkages in the lignin-carbohydrate complex (LCC) could explain the better performance of a higher OA/ChCl ratio in biomass deconstruction.[46,47] Particularly, carboxylic-containing HBDs or Brønsted acids such as oxalic acid and lactic acid at higher mole fractions could donate more active protons for promoting the catalytic reaction.[28,48] In addition to the viscosity, the polarity of solvents is another significant factor relating to the ability of solvents to solubilize the biomass components. In this study, the polarity of NADESs was experimentally determined using Kamlet−Taft polarity scales (as described in Supporting Information). Table S3 shows that the mixture with a larger OA/ChCl ratio tended to have a higher value of the net hydrogen bond acidity (or the $\alpha-\beta$ value), indicating a greater acidic character. Among 12 NADESs, DES-06 had the highest $\alpha-\beta$ value, correlated well with its most efficiency in hemicellulose removal (58.9%) and delignification (38.8%).

Besides the effect of the HBD/HBA ratio, Figure 1 also demonstrates that the degree of hydration in NADESs influenced the pretreatment efficiency. Even though the increase in water mole fraction could significantly reduce the viscosity of NADESs that was possible to increase the mass transfer coefficient, the results demonstrated that higher water contents in NADESs in turn decreased hemicellulose removal and delignification. The xylan and lignin losses gradually dropped from 45.1 to 27.9% and 29.9 to 20.9%, respectively, with increasing water in ChCl/OA (2:1) from 0% (DES-12) to

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Table 2. Chemical Compositions of CP After the MW-NADES Process at 100 °C for 15 min

		chemical composition (wt %)[a]			
pretreatments	solid recovery (%)	glucan	xylan	AIL[b]	ASL[c]
untreated	100.00	31.88 ± 0.29	15.83 ± 0.93	29.77 ± 0.31	4.10 ± 0.01
DES-01	94.06	27.15 ± 0.42	12.14 ± 0.39	25.37 ± 0.02	3.13 ± 0.03
DES-02	89.13	30.53 ± 0.14	10.57 ± 0.28	25.76 ± 0.17	2.79 ± 0.02
DES-03	87.55	31.05 ± 0.13	10.63 ± 0.19	24.90 ± 0.14	2.78 ± 0.01
DES-04	89.21	30.88 ± 0.16	11.67 ± 0.38	24.76 ± 0.20	3.05 ± 0.02
DES-05	86.04	32.65 ± 1.13	11.61 ± 0.57	24.26 ± 0.15	2.81 ± 0.00
DES-06	78.50	35.52 ± 0.34	8.29 ± 0.76	23.53 ± 0.17	2.85 ± 0.03
DES-07	89.60	32.54 ± 0.21	10.67 ± 0.35	24.86 ± 0.15	3.08 ± 0.10
DES-08	89.23	31.10 ± 0.84	10.38 ± 0.88	25.55 ± 0.09	3.00 ± 0.01
DES-09	82.45	33.56 ± 0.52	8.94 ± 0.62	23.45 ± 0.21	3.31 ± 0.03
DES-10	96.53	27.44 ± 0.43	11.98 ± 0.38	25.41 ± 0.11	3.39 ± 0.00
DES-11	85.88	32.99 ± 0.41	9.77 ± 0.73	23.65 ± 0.37	3.05 ± 0.01
DES-12	85.72	32.78 ± 0.17	10.13 ± 0.56	24.58 ± 0.15	3.12 ± 0.02

[a]The chemical compositions are presented as g per 100 g pretreated samples (as dry basis). [b]AIL: acid-insoluble lignin (Klason lignin). [c]ASL: acid-soluble lignin.

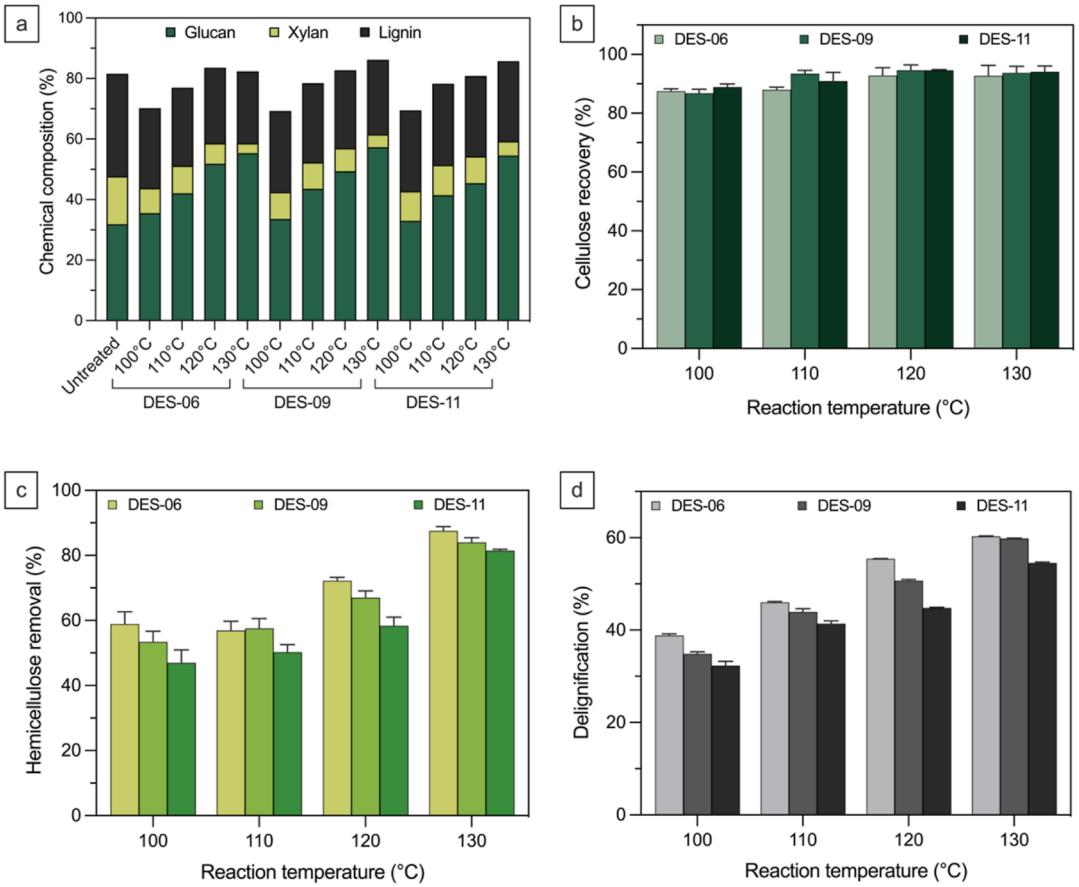


Figure 2. Pretreatment efficiency of selected NADESs at different microwave temperatures (100 to 130 °C). (a) Chemical compositions of untreated and DES-pretreated CPs. Percentage of (b) cellulose recovery, (c) hemicellulose removal, and (d) delignification, after the MW-NADES pretreatment at various microwave temperatures relative to untreated CP.

85 mol % (DES-01). These findings were attributed to the substantial disruption of DES structure and the lower acidity of NADESs by higher levels of water addition, thus reducing the performance of NADESs in biomass fractionation. In contrast, the results indicated that the very low water content in NADESs had a favorable effect on pretreatment efficiency. For ChCl/OA (2:1), the higher removals of hemicellulose (47.0%) and lignin (32.3%) were achieved by the mixture containing 10 mol % water (DES-11), compared to the dry one (DES-12) and the mixture with higher water contents. Additionally, it was found that the small amount of incorporated water could enhance the acidity of NADESs. As a result, the present study showed that the NADESs based on the ChCl−OA−water system could be highly tunable by adjusting their molar compositions. Our results suggest that the NADES with the

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highest acidic character of NADESs plays a critical role in biomass deconstruction.

According to the results, the pretreatment using NADESs based on the ChCl−OA−water system mostly affected hemicellulose and lignin dissociation rather than cellulose solubilization. The untreated CP mainly comprised 31.9% of cellulose, 15.8% of hemicellulose, and 33.9% of lignin. It was found that all NADESs had the ability to solubilize cellulose in CP less than 20%. Therefore, the lower yield of solid recovery mainly contributed to hemicellulose and lignin solubilization (Table 2). Among 12 NADESs, the minimum solid recovery (78.5 wt %) was obtained from the DES-06 pretreatment, corresponding to the greatest efficiency of DES-06 in removing hemicellulose and lignin. Moreover, the highest cellulose content was observed in the DES-06 pretreated fiber (35.5%), followed by DES-09 pretreated fiber (33.6%) and DES-11 pretreated fiber (33.0%). Consequently, based on three key performance indicators (cellulose content, hemicellulose removal, and delignification), DES-06, DES-09, and DES-11 were selected as pretreatment solvents for further investigation.

Effect of Reaction Temperature in Microwave Processing on Fractionation Efficiency of the MW-NADES Pretreatment. In this experiment, the fractionation efficiency of the three selected NADESs (DES-06, DES-09, and DES-11) in the MW-NADES process was investigated at different reaction temperatures (100 to 130 °C). The results demonstrated that the increment of microwave temperature could remarkably enhance the removal rates of hemicellulose and lignin for all three solvents but slightly affect the cellulose solubilization (Figure 2). The corresponding yields of solid recovery were lowered as the microwave temperature increased (Table S4). It could be seen that the improved fractionation efficiency of NADESs was in the following order: DES-06 > DES-09 > DES-11. Figure 2c,d shows that increasing microwave temperature from 100 to 130 °C in DES-06 pretreatment could increase the hemicellulose removal from 58.9 to 87.5% and improve the degree of delignification from 38.8 to 60.3%. The increased biomass fractionation could be explained by temperature-induced diffusion. As the temperature increased, the solvent viscosity was significantly reduced (Figure S2), enhancing solvent accessibility to the recalcitrant structure of LCBs and driving the mass transfer process during microwave-assisted extraction. Consequently, this process could increase the intracellular pressure leading to promoting the cell rupture.[49,50] In addition, the recovery yields of cellulose maintained over 80% (relative to the untreated CP) over the reaction temperature range (Figure 2b), implying that the increased temperatures negligibly disrupted the cellulose structure. As can be seen in Figure 2a, the hemicellulose and lignin contents dropped to 3.3 and 23.8%, respectively, while the cellulose content was enriched to 55.4% after the DES-06 pretreatment at the highest temperature. It is noticed that the mass balance for all pretreatments was not close to 100%, although soluble extractives representing 6−11 wt % of the pretreated sample were included. The lower mass balance closure was observed in the biomass samples pretreated at lower microwave temperatures. These mild conditions resulted in the low ability of NADESs to solubilize lignocellulosic components of CP as well as other biomass components that were unaccounted for in this analysis. As reported by Mendoza Martinez et al., CP was chemically composed of extractives (21.95%), uronic acid (5.50%), acetic acid (2.85%), arabinan (3.45%), galactan (1.65%), and mannan (1.00%) besides lignocellulosic components (29.60% of glucan, 14.55% of xylan, and 26.59% of lignin).[11] Based on the results, although the increase in reaction temperature in the MW-NADES process could boost the cellulose content and the hemicellulose removal, lignin was partially removed. The insufficient delignification was due to the highly lignified cell walls of CP that limited the biomass accessibility to solvents.

Fractionation Efficiency of Sequential Pretreatment Using MW-NADES Followed by AHP. Based on the MW-NADES pretreatment, DES-06 exhibited the most powerful solvent for CP fractionation with high hemicellulose removal and partial delignification. The AHP bleaching is considered a greener oxidative process alternative to sodium chlorite bleaching and has been reported to remove lignin effectively. Under alkaline conditions, H_2O_2 dissociated to generate the strong oxidants, including hydroperoxyl anion (HOO^-), hydroxyl radical ($HO^\bullet$), and superoxide anion radical ($O_2^{\bullet-}$), that are responsible for promoting delignification process and biomass depolymerization.[31,51,52] Our preliminary study revealed that only a single-step AHP process could remove almost half of the lignin from CP even at higher H_2O_2 concentration (5% w/v or 1000 mg/g dried CP), indicating insufficient delignification to obtain CRM. Several publications reported that the delignification could synergistically be improved when the AHP process was assigned as a second stage of other pretreatments such as DES (ChCl/lactic acid),[53] diluted acid (3% sulfuric acid),[54] and organosolv process.[55]

In this study, Figure 3b demonstrates that the sequential MW-NADES and AHP pretreatment could remove hemicellulose and lignin from CP extensively. In particular, the fractionation efficiency of the sequential pretreatment was increased following the increment of microwave temperature in the MW-NADES pretreatment. The results demonstrated that the DES-06 pretreatment at 130 °C, followed by a single-step AHP process, showed the highest removals of hemicellulose (94.6%) and lignin (94.7%) from CP, and cellulose recovery of 82.3%, compared to other NADESs and lower temperatures. Subsequently, the proportion of cellulose in the pretreated fibers was enriched to 81.1%, as shown in Figure 3a. It might be explained that microwave processing at higher operating temperatures could greatly assist the NADESs in breaking the recalcitrant cell wall of CP. Eventually, the disrupted biomass facilitated the strong oxidant accessibility to promote the cleavage of LCC linkages.

Furthermore, a higher cellulose purity was expected prior to nanocellulose production. It was found that the double-step AHP process (dAHP) could intensify biomass depolymerization. Figure 3b also illustrates that after repeating the AHP process, the maximum removals of hemicellulose and lignin could be achieved at 99.6 and 97.7%, respectively. Unfortunately, the sequential DES-dAHP pretreatment could remarkably promote cellulose depolymerization, resulting in 68.6% cellulose recovery. Regardless of the recovery yields, the DES-dAHP pretreatment could increase the cellulose content to 86.1% in the pretreated fiber at the end.

Recently, Huang et al.[54] performed the sequential pretreatment on poplar sawdust using diluted acid (3% sulfuric acid) at 180 °C for 1 h, followed by AHP (5% w/v H_2O_2) at 80 °C for 1 h. The results showed that the diluted acid pretreatment could remove 92.0% of hemicellulose but only 8.2% of lignin from poplar sawdust. However, the sequential pretreatment could increase the delignification to 63.6%. In the present study, the pretreatment using acidic DESs under microwave

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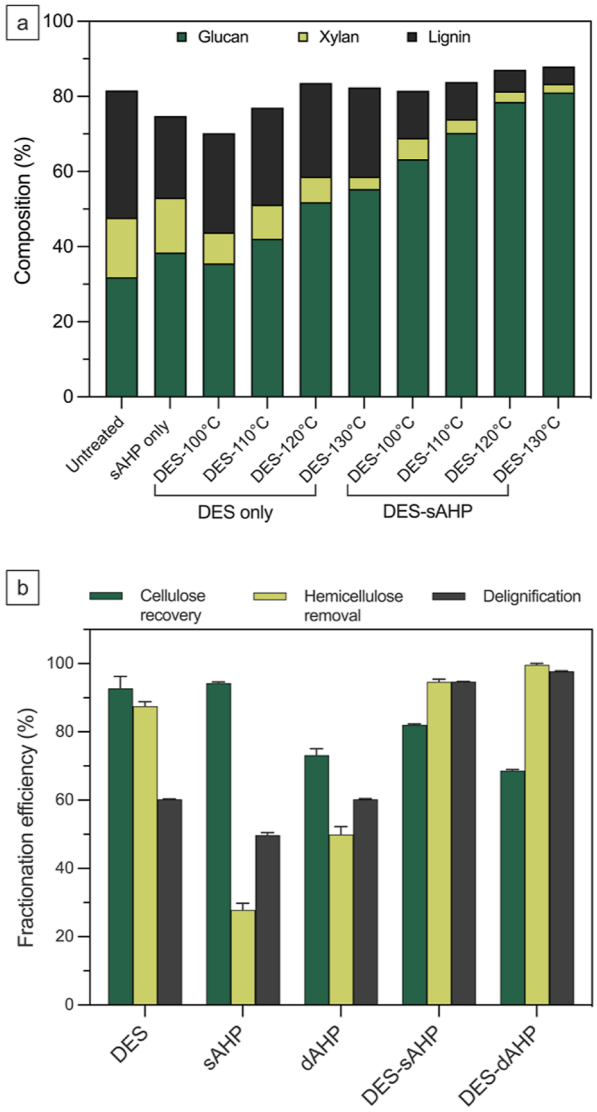


Figure 3. Composition change and pretreatment efficiency upon pretreatments. (a) Chemical compositions of untreated and pretreated CPs: DES represents the DES-06 pretreatment at various microwave temperatures, sAHP represents the single-step AHP process, and DES-AHP represents the sequential pretreatment using DES followed by AHP. (b) Fractionation efficiency of different pretreatment approaches: DES represents the DES-06 pretreatment at 130 °C and dAHP represents the double-step AHP process.

irradiation was effective in eliminating hemicellulose (>80%) and partially removing lignin (>50%) from CP. Accordingly, the MW-NADES pretreatment may be used for biomass fractionation alternative to conventional inorganic acid pretreatment due to its high ability to remove noncellulosic parts, the use of environmentally friendly solvents, and mild operational conditions (lower reaction temperature and shorter processing time). Additionally, the sequential MW-NADES and AHP pretreatment was a sustainable and efficient approach providing synergistic effects on biomass fractionation at which noncellulosic compositions were almost completely improved, leading to producing cellulose-rich fibers. However, there were some limitations to our sequential pretreatment. The AHP process in this study required a long processing time; thus, the process optimization of AHP pretreatment, including varying temperature, time, and H_2O_2 concentration, needed to be performed in further investigation.

Impact of Pretreatment Process on Morphological Characteristics of CP. The SEM images revealed changes in the morphology of CP upon the pretreatments, as shown in Figure 4. The untreated CP showed bundles of microfibrils with a smooth surface, indicating the presence of thickened lignified cell walls in CP. After the MW-NADES pretreatment, a rough fractured feature was presented on the surface of DES-CP. This morphological alteration might result from the hemicellulose dissociation and partial removal of lignin from CP via microwave-induced cell disruption.[56] Additionally, the microfibril bundles were found to be disassembled and separated into individual fibers after the DES-CP was exposed to the AHP solution. The significant surface changes of DES-AHP CP were due to the increased solubilization of noncellulosic components from CP. The higher degree of biomass deconstruction after the sequential pretreatment might be associated with the increased accessible surface area of biomass through microwave processing, facilitating the penetration of strong oxidants to cleave the LCC linkages.[57−59]

Impact of Pretreatment Process on Chemical Structure of CP. The ATR-FTIR spectra of untreated and pretreated CP are illustrated in Figure S3. The spectral range was magnified to 2000−500 cm^{-1} to facilitate the identification of functional groups and fingerprint region, as presented in Figure 5a. According to the spectral analysis, it could be verified that the sequential pretreatment could successfully remove hemicellulose and lignin from CP, while the cellulose fraction was highly maintained. The FT-IR spectra for the untreated and pretreated samples had very similar peaks at 3332 cm^{-1} (O−H stretching vibration) and in the 2900 cm^{-1} region (aliphatic C−H stretching vibration). The strong absorption peak at 1732 cm^{-1} in untreated CP corresponding to the C═O stretching vibration mode of ester groups in hemicellulose was depressed in the spectrum for DES-06 and disappeared in the DES-AHP spectrum, reflecting a significant removal of hemicellulose during the MW-NADES pretreatment and nearly complete loss after the sequential pretreatment.[60] The vibration bands at 1643 and 1616 cm^{-1} in the spectra of CNC were attributed to the O−H bending of absorbed water.[61] The characteristic bands associated with lignin at 1594 cm^{-1} (aromatic skeletal vibrations and C═O stretching), 1506 cm^{-1} (aromatic skeletal vibrations), 1454 cm^{-1} (C−H deformations), and 1236 cm^{-1} (C−O stretching in syringyl ring) were all presented in the untreated CP spectrum.[62−64] Some of the distinctive peaks were still expressed in the spectra for DES-CP but no longer observed in the spectrum for DES-AHP CP, indicating the substantial deconstruction of the lignin structure after the sequential DES-AHP process. However, the peak of the C−H in-plane deformation in lignin at 1423 cm^{-1} existed in all spectra due to the overlapping with CH_2 symmetric bending in cellulose. Additionally, the intensities of the bands located at 1369 cm^{-1} (C−H bending in cellulose), 1315 cm^{-1} (CH_2 wagging in cellulose), and 1157 cm^{-1} (C−O stretching in cellulose and C−O−C asymmetrical stretching in xylan) were higher for all pretreated samples compared with untreated CP. The strong absorption band between 1105 and 899 cm^{-1} with the maximum at 1024 cm^{-1} indicated cellulose glycosidic linkages.[64,65] Furthermore, the FT-IR spectrum for the DES-AHP-pretreated sample overall was similar to the CNC spectrum. These findings confirmed the removal of hemicellulose and

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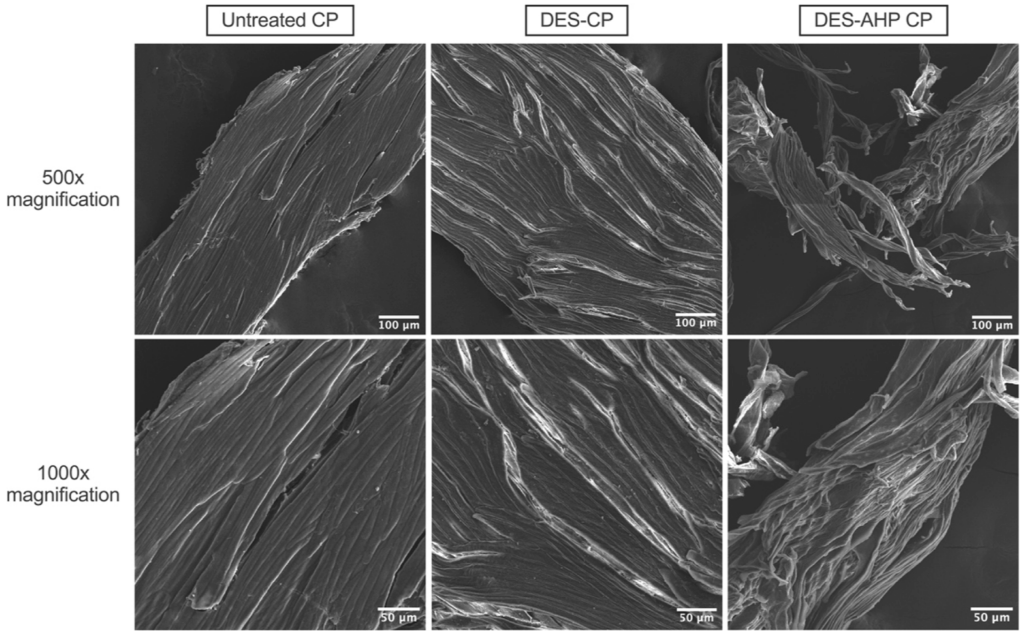


Figure 4. SEM images of untreated, DES-pretreated, and DES-AHP-pretreated fibers at 500× magnification (top), and 1000× magnification (below).

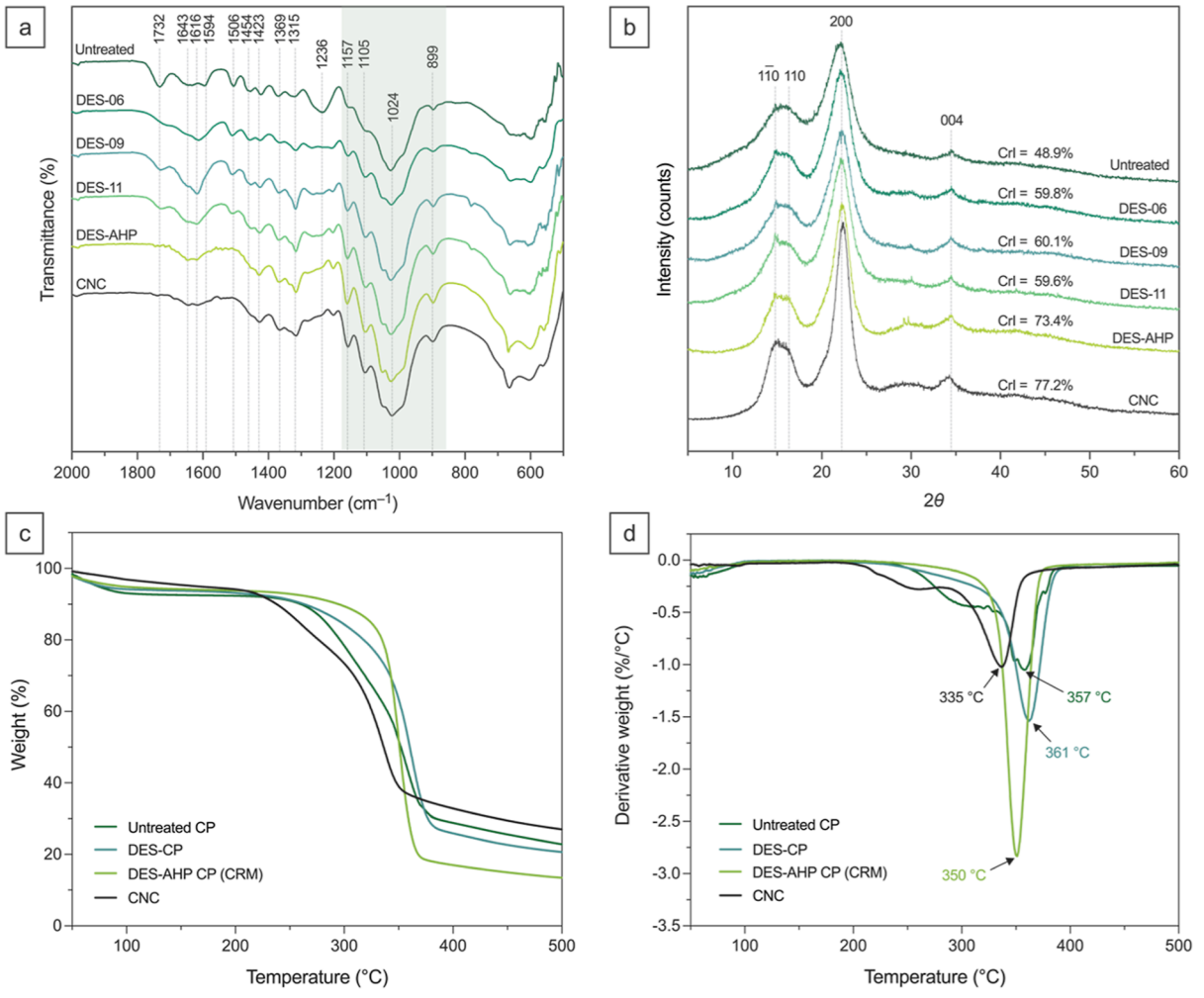


Figure 5. Impact of pretreatment process on physicochemical properties of CP. (a) Magnified ATR-FTIR spectra, (b) XRD patterns, (c) thermogravimetric curves, and (d) derivative thermogravimetric curves for untreated and pretreated CPs.

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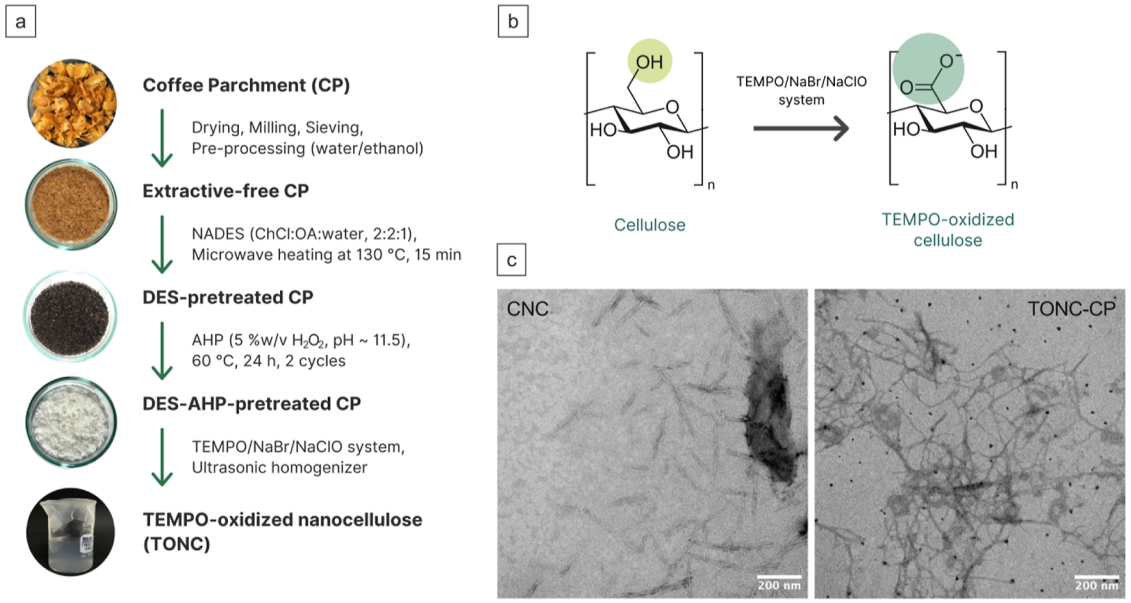


Figure 6. Nanocellulose production. (a) Schematic representation of the sequential pretreatment process for the isolation of cellulose-rich fibers from CP and the preparation of nanocellulose. (b) Scheme of TEMPO-mediated oxidation. (c) TEM images of commercial CNCs (left) and TONC-CP (right) at 100,000× magnification.

lignin from CP and the enrichment of cellulose fraction in pretreated fibers after the sequential pretreatment. The results of the FT-IR analysis agreed well with the compositional analysis.

Impact of Pretreatment Process on Crystalline Structure of CP. The XRD patterns for untreated CP and pretreated samples are presented in Figure 5b. The diffractograms reveal that pretreatment processes could enhance the crystalline regions of CP. The characteristic peaks of cellulose I at 2θ = 14.6, 16.4, 22.6, and 34.5°, corresponding to the $1\bar{1}0$, 110, 200, and 004 crystallographic planes, respectively,[33,66] were observed in all samples. Based on Segal's equation, the CrI values of CP increased from 48.9% (untreated CP) to around 59.8% after the MW-NADES pretreatment. Interestingly, the sequential DES-AHP pretreatment could increase the CrI value of CP up to 73.4%, which was close to the CrI value of commercial CNC (77.2%). The increase in crystallinity was attributed to the destruction of the amorphous structure (hemicellulose−lignin complex) in biomass.[67,68] The results proved that the sequential pretreatment could effectively remove the amorphous matrix of hemicellulose and lignin and maintain the crystalline structure of cellulose in pretreated fibers.

Thermal Stability of Pretreated CP. The thermal decomposition behaviors and thermal stability of untreated and pretreated CPs were evaluated from thermogravimetric (TG) and derivative thermogravimetric (DTG) curves, as shown in Figure 5c,d. The results exhibited that the CRM obtained from the sequential pretreatment of CP showed better thermal stability than untreated CP due to its higher onset decomposition temperature. The TGA curves reveal that the initial mass loss of all samples occurring below 200 °C might be mainly contributed to the evaporation of absorbed water and light components from the biomass.[69] The onset temperature of the primary devolatilization process of CP increased from around 275 °C to above 300 °C after the MW-NADES process and up to nearly 350 °C after the sequential pretreatment. In addition, the TGA curves also show that the pretreated fibers had a higher onset decomposition temperature than commercial CNC (ca. 225 °C). These findings indicated that the obtained cellulosic fibers (or DES-AHP CP) had higher thermal stability than untreated CP and commercial nanocellulose. The devolatilization temperature of untreated and pretreated CPs could extend to 400 °C. In this interval, the lignocellulosic components are mostly ignited to release the volatile matter, which is also called as pyrolysis process due to the absence of oxygen.[70,71]

As observed from the DTG curves, the strong peaks referred to the maximum weight loss rates at certain pyrolysis temperatures (T_{max}), caused by the devolatilization process of biomass. The DTG curve of untreated CP displays two distinct decomposition peaks over the temperature range; a shoulder peak at a lower temperature (ca. 300 °C) contributed to the xylan (hemicellulose) degradation, and a stronger peak at 357 °C was mainly ascribed to the cellulose decomposition.[70,72] After the pretreatment processes, the shoulder peak was found to be disappeared, but a single pyrolysis peak at 361 °C (for DES-CP) and 350 °C (for DES-AHP CP) was still visible. The results indicated evidence of the efficient hemicellulose removal and the cellulose remaining during the MW-NADES process and sequential DES-AHP pretreatment.[73] However, it was noted that the pyrolysis of lignin proceeded in a wide range of temperatures (200 to 600 °C), which overlapped the hemicellulose and cellulose decomposition temperatures.[74] It has been reported that the lowered decomposition temperature with two-step degradation of CNCs was associated with the presence of sulfate half-ester groups and the smaller fiber dimensions.[75] In the third stage, at above 400 °C, all the samples revealed a slow and continuous weight loss, and the main residues were the fixed carbon.[76]

CP-Derived Nanocellulose. The schematic diagram of the nanocellulose production from cellulosic fibers extracted from CP via the proposed sequential pretreatment is illustrated in Figure 6a. It could be seen that after the MW-NADES pretreatment, the physical appearance of CP became a darker shade of brown. However, AHP could enhance brightness for

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DES-pretreated CP, making the cellulosic fibers (or CRM) more favorable for nanocellulose production. The surface modification of CRM, followed by ultrasonication, could produce a translucent viscous liquid called TONC-CP, inferring a partial disintegration of cellulose fibers.[77] The DLS analysis demonstrates that the TONC-CP had an average particle size of 115.2 ± 1.5 nm and a zeta potential value of −52.5 ± 1.6 mV, indicating nanosized particles with good colloidal stability (Figures S4). The TEMPO-mediated oxidation is a reaction for selective conversion of the C-6 carbon hydroxyl group of cellulose to the carboxylic group using TEMPO/NaBr/NaClO system (Figure 6b).[78] The successful oxidation of cellulose was confirmed by the presence of a strong absorption peak at 1603 cm^{-1} (corresponding to the C═O stretching in carboxyl groups) in the ATR-FTIR spectra of TONC-CP (Figure S5).[79,80] As shown in Figure 6c, TEM images revealed that the prepared TONC-CP possessed a fiber-like network structure with a cross-sectional dimension of 3−10 nm, which was smaller than commercial CNCs (3−12 nm). These findings suggested that the production process could effectively transform the cellulose fibers isolated from CP into cellulose nanoparticles.

CONCLUSIONS

Our findings showed that the cellulosic fibers could be effectively isolated from CP by the sequential pretreatment using NADES (ChCl/OA/water in a ratio of 2:2:1) with microwave heating at 130 °C for 15 min, followed by a two-step AHP process. This pretreatment process could almost completely remove hemicellulose (99.6%) and lignin (97.7%) from CP. As a result, the cellulose content was enriched to 86.1% in the pretreated fibers. The obtained cellulosic fibers (or CRM) exhibited increased crystallinity with higher thermal stability. Furthermore, the TEMPO-mediated oxidation, followed by ultrasonic homogenization, effectively transformed the CRM into very fine fiber-like nanoparticles. Therefore, the proposed sequential pretreatment revealed an efficient and sustainable approach to valorize the agricultural residues for the production of nanocellulose, which would be a promising renewable nanomaterial for various biomedical and pharmaceutical applications.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.3c02679.

Phase diagram of the ChCl−OA−water system, additional experimental methods and results for measurements of viscosity and Kamlet−Taft parameters of the studied NADESs, solid residues of DES-pretreated CPs, ATR-FTIR spectra for pretreated fibers, experimental details and results for particle size analysis of TONC, and ATR-FTIR spectra for TONC (PDF)

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Notes

The authors declare no competing financial interest.

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